

SADDLEPOINT APPROXIMATIONS FOR THE DISTRIBUTION OF REGRESSION QUANTILES

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Summary: Since their introduction in 1978, regression quantiles have played an increasingly important role in regression analysis. In particular, they form the basis for quantile regression, a statistical technique intended to draw inferences about conditional quantile functions. Although it is well known that regression quantiles have limiting normal distributions, not much is known about their behaviour in finite and small samples. In the present paper we study this problem and, in particular, apply the saddlepoint method to obtain improved small sample approximations. These approximations are considered in a numerical study where they are found to perform quite well even in fairly small samples.

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1. Introduction

Regression quantiles (RQs) were introduced in 1978 by Koenker and Bassett as natural analogues of order statistics in the regression situation. They also proposed them as the basis for defining regression trimmed means and, more generally, regression L-statistics. Since the original paper by Koenker and Bassett, a large number of papers have appeared on the theory and application of RQs, extending it in many directions. RQs also (naturally) form the basis of quantile regression – a statistical technique intended to estimate and draw inferences about conditional quantile functions. It is well known that RQs have limiting normal distributions, in form closely related to those of ordinary sample quantiles (see e.g. Koenker and Bassett, 1978, Ruppert and Carroll, 1980). However, not much is known about the behaviour of RQs in finite and small sample sizes. In the present paper we study this problem and, in particular, apply the saddlepoint method to obtain small sample approximations to the distribution of RQs.

The saddlepoint method provides good approximations to, *inter alia*, small tail probabilities or to the density in the tails of distributions. Since its introduction into statistics in the seminal paper by Daniels (1954) it has been used successfully in a variety of situations, (see e.g. Reid, 1988, 1995). In Daniels (1983), the saddlepoint method was extended to statistics defined via estimating equations and based on i.i.d. observations. In Strawderman, Casella and Wells (1996) this was extended to the independent, but not necessarily identically distributed, case. This case is of particular interest in the regression situation. A related paper is that of Spady (1991) who extended the Daniels result to the regression situation, obtaining an approximation different

to that of Strawderman et al. (1996). As a special case, he applied his results to L_1 regression. Ronchetti and Welsh (1994) obtained saddlepoint approximations for multivariate M-estimators, which again are applicable to the regression case. We use these results to obtain saddlepoint approximations to the distribution of regression quantiles. These we compare to the exact distribution, estimated via Monte Carlo simulation, as well as to the asymptotic normal approximation.

The layout of the paper is as follows. In the next section, we give a brief overview of RQs and their limiting distribution theory. In Section 3 we derive the saddlepoint approximation for the RQs. Section 4 describes a simulation study that was carried out in order to assess the performance of the approximations. The results of that study are given in section 5 and discussed in section 6. The results indicate improvements of the saddlepoint approximation over the normal approximation in most cases with substantial improvements in a number of them. The final section contains conclusions and directions for some further possible research.

2. Regression Quantiles

Let Y be a random variable with distribution function F and define for $0 < \tau < 1$ the quantile function in the usual way by

$$Q(\tau) = \inf \{y : F(y) \geq \tau\}. \quad (1)$$

The latter may be formulated as the solution to an optimization problem. Define the piecewise linear check function ρ_τ by

$$\rho_\tau(u) = u(\tau - I(u < 0)). \quad (2)$$

Minimizing the expectation of $\rho_\tau(Y - \xi)$ with respect to ξ yields solutions, the smallest of which is $Q(\tau)$. The sample analogue of $Q(\tau)$, based on a

sample $Y_1, \dots, Y_n$ on Y , is the τ -th sample quantile, found by solving

$$\min_{\xi \in R} \sum_{i=1}^n \rho_{\tau}(Y_i - \xi). \quad (3)$$

This formulation of sample quantiles (not using an ordering of the sample) yields a natural generalization to the regression case. This will now be discussed.

Consider the usual linear model

$$Y_i = \underline{x}'_i \underline{\beta} + e_i, \quad i = 1, 2, \dots, n \quad (4)$$

with $\{e_i\}$ i.i.d. errors having distribution function F and

$$X = \begin{bmatrix} \underline{x}'_1 \\ \underline{x}'_2 \\ \vdots \\ \underline{x}'_n \end{bmatrix} \quad (5)$$

an $n \times p$ design matrix containing an intercept term. Here $\underline{\beta}$ is a $p \times 1$ vector of regression parameters. For any $\underline{b} \in R^p$, let

$$r_i \equiv r_i(\underline{b}) = Y_i - \underline{x}'_i \underline{b}, \quad i = 1, 2, \dots, n \quad (6)$$

denote the corresponding residuals. By analogy to the above, define the τ -th regression quantile as

$$\widehat{\underline{\beta}}_{RQ}(\tau) = \arg \min_{\underline{b}} \sum_{i=1}^n \rho_{\tau}(r_i(\underline{b})). \quad (7)$$

Koenker and Bassett (1978) and Ruppert and Carroll (1980) showed that the behaviour of $\widehat{\underline{\beta}}_{RQ}(\tau)$ is analogous to that of the sample quantiles. In particular, they showed that

$$\sqrt{n} \left(\widehat{\underline{\beta}}_{RQ}(\tau) - \underline{\beta}(\tau) \right) \xrightarrow{D} N(\underline{0}, \sigma^2(\tau) \Omega^{-1}), \quad (8)$$

as $n \rightarrow \infty$, where

$$\sigma^2(\tau) = \frac{\tau(1-\tau)}{f(Q(\tau))^2}, \quad (9)$$

with $f = F'$ and

$$\Omega = \lim_{n \rightarrow \infty} n^{-1} (X'X). \quad (10)$$

Note that we assume the required conditions to hold. Here $\underline{\beta}(\tau)$ is the parameter being estimated by $\widehat{\underline{\beta}}_{RQ}(\tau)$ and is given by

$$\underline{\beta}(\tau) = \underline{\beta} + Q(\tau) \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}. \quad (11)$$

Our interest lies in the accuracy of the above normal approximation and how it can be improved using a saddlepoint technique. In the next section we show how the saddlepoint approximation can be applied to RQs.

3. Saddlepoint approximation for regression quantiles

Although saddlepoint approximations for statistics based on i.i.d. observations have been studied for many years, in the regression case they have only been studied during recent years. An important contribution was that of Strawderman et al. (1996) where they extended the results of Field (1982) to estimating equations that can be expressed as sums of independent, but not necessarily identically distributed, random variables. This applies in particular to regression estimators that are obtained as solutions to estimating equations. Results specifically for the regression case were obtained earlier, but by a different approach, by Spady (1991). He applied his results to the special case of L_1 regression and demonstrated the accuracy of the approximation in a case where the usual normal approximation is very poor. Another approximation for the regression case was obtained by Ronchetti and Welsh (1994), as a special case of multivariate M-estimators. Their results also apply

to our situation because RQs are also M-estimators. Although these authors obtain (apparently) different approximations, using different methods, they give essentially the same numerical results in the specific cases on which we tested them.

We found the results of Spady (1991) to be most convenient for our application. In particular, we will apply his expression (2.8) to the case of regression quantiles. In this section we derive this expression for the density of a regression quantile, using to a large extent his notation.

Consider again the regression model and RQs as given in the previous section. For ρ_τ defined by (2), let

$$\psi_\tau(u) = \tau - I(u < 0) \quad (u \neq 0) \quad (12)$$

denote its directional derivative. The estimator $\widehat{\beta}_{RQ}(\tau)$ is given as a solution to the estimating equations, defined by the directional derivatives

$$\psi_j(\widehat{\beta}_{RQ}(\tau)) = \sum_{i=1}^n \psi_\tau(Y_i - \underline{x}_i' \widehat{\beta}_{RQ}(\tau)) \cdot x_{ij}, \quad j = 1, 2, \dots, p \quad (13)$$

where x_{ij} is the j -th element of the vector $\underline{x}_i$. (See Spady, 1991, section 3 and also Daniels, 1983, p. 92, 93, where a similar approach is applied to the usual sample quantiles.)

Writing

$$\psi_{ij}(b) = \psi_\tau(Y_i - \underline{x}_i' b) x_{ij}, \quad (14)$$

the Ψ -function associated with observation i is given by

$$\begin{aligned}\underline{\psi}_i(\underline{\hat{\beta}}_{RQ}(\tau)) &\equiv \begin{bmatrix} \psi_{i1}(\underline{\hat{\beta}}_{RQ}(\tau)) \\ \vdots \\ \psi_{ip}(\underline{\hat{\beta}}_{RQ}(\tau)) \end{bmatrix} \\ &= \begin{bmatrix} \psi_\tau(Y_i - \underline{x}'_i \underline{\hat{\beta}}_{RQ}(\tau)) x_{i1} \\ \vdots \\ \psi_\tau(Y_i - \underline{x}'_i \underline{\hat{\beta}}_{RQ}(\tau)) x_{ip} \end{bmatrix} \\ &= \Psi_\tau(Y_i - \underline{x}'_i \underline{\hat{\beta}}_{RQ}(\tau)) \underline{x}_i.\end{aligned}\quad (15)$$

Note that with $\underline{\beta}(\tau)$ as in (11), it follows by direct calculation that

$$E\underline{\psi}_i(\underline{\beta}(\tau)) = \underline{0}, \quad i = 1, 2, \dots, n. \quad (16)$$

Now, let $\underline{b}$ be fixed. In order to derive the saddlepoint approximation of $\underline{\hat{\beta}}_{RQ}(\tau)$, i.e. (2.8) in Spady (1991), we need to find the appropriate saddlepoint by solving the set of equations

$$\underline{K}'(\underline{t}) = \underline{0} \quad (17)$$

where

$$\begin{aligned}K(\underline{t}) &\equiv K(\underline{t}, \underline{b}, \underline{\beta}) \\ &\equiv \log \prod_{i=1}^n M_i(\underline{t}, \underline{b}, \underline{\beta}) \\ &= \sum_{i=1}^n \log M_i(\underline{t}, \underline{b}, \underline{\beta}).\end{aligned}\quad (18)$$

Here M_i is the moment generating function (m.g.f.) of $\underline{\psi}_i$ and by independence $K(\underline{t})$ is thus the joint cumulant generating function of $(\underline{\psi}_1, \dots, \underline{\psi}_n)$.

Now, from (15) and (4) we have

$$\begin{aligned}
 M_i(\underline{t}) &\equiv M_i(\underline{t}, \underline{b}, \underline{\beta}) = E e^{\underline{t}' \underline{\psi}_i(\underline{b})} \\
 &= E \exp [\Psi_\tau(Y_i - \underline{x}'_i \underline{b}) \underline{x}'_i \underline{t}] \\
 &= E \exp [\Psi_\tau(e_i - \underline{x}'_i(\underline{b} - \underline{\beta})) \underline{x}'_i \underline{t}] \\
 &= \int_{-\infty}^{\infty} \exp [\Psi_\tau(u - \underline{x}'_i(\underline{b} - \underline{\beta})) \underline{x}'_i \underline{t}] dF(u)
 \end{aligned}$$

since the $\{e_i\}$ are i.i.d. F .

Using the definition of Ψ_τ it follows easily from the above that

$$\begin{aligned}
 M_i(\underline{t}) &= e^{(\tau-1)\underline{x}'_i \underline{t}} F(\underline{x}'_i(\underline{b} - \underline{\beta})) + e^{\tau \underline{x}'_i \underline{t}} (1 - F(\underline{x}'_i(\underline{b} - \underline{\beta}))) \\
 &= e^{(\tau-1)\underline{x}'_i \underline{t}} \left[F(\underline{x}'_i(\underline{b} - \underline{\beta})) + e^{\underline{x}'_i \underline{t}} (1 - F(\underline{x}'_i(\underline{b} - \underline{\beta}))) \right] \quad (19)
 \end{aligned}$$

Using this in (18) gives

$$K(\underline{t}) = \sum_{i=1}^n \left[(\tau-1) \underline{x}'_i \underline{t} + \log \left(F_i + e^{\underline{x}'_i \underline{t}} (1 - F_i) \right) \right] \quad (20)$$

where we use the shorthand notation

$$F_i = F(\underline{x}'_i(\underline{b} - \underline{\beta})). \quad (21)$$

Differentiating (20) with respect to the components of $\underline{t}$ gives a vector of derivatives

$$\underline{K}'(\underline{t}) = \sum_{i=1}^n \underline{x}'_i \left[F_i + e^{\underline{x}'_i \underline{t}} (1 - F_i) \right]^{-1} \left[(\tau-1) F_i + \tau e^{\underline{x}'_i \underline{t}} (1 - F_i) \right]. \quad (22)$$

The saddlepoint, denoted by $\underline{T}_0$, is obtained using (22) in solving the saddlepoint equation (17).

Furthermore, in applying (2.8) of Spady (1991), we also need the matrix of second derivatives of $K(\underline{t})$ (to be evaluated at $\underline{T}_0$). Differentiating (22)

again and simplifying, gives the matrix

$$K''(t) = \sum_{i=1}^n \underline{x}_i \underline{x}_i' e^{\underline{x}_i' t} F_i (1 - F_i) \left[F_i + e^{\underline{x}_i' t} (1 - F_i) \right]^{-2}. \quad (23)$$

Finally, we need the last term in Spady's (2.8). For this, proceed as in his (3.3). This needs some careful handling of the value of the directional derivative (12) when evaluated at 0.

Now, from the definition of $\underline{\psi}_i$ as a function of an indicator function and remembering that the derivative of the Heaviside (indicator) function is given by the Dirac delta function δ (see e.g. Wilmott et al., 1995, p. 63), it follows that for $i = 1, 2, \dots, n$

$$\frac{\partial \underline{\psi}_i}{\partial \underline{b}} = -\underline{x}_i \underline{x}_i' \delta(Y_i - \underline{x}_i' \underline{b}). \quad (24)$$

Thus, in Spady's notation

$$\begin{aligned} E_g \left[\frac{\partial \underline{\psi}_i}{\partial \underline{b}} \right] &= -\underline{x}_i \underline{x}_i' \int_{-\infty}^{\infty} \delta(u - \underline{x}_i'(\underline{b} - \underline{\beta})) e^{T_0' \underline{\psi}_i(u - \underline{x}_i'(\underline{b} - \underline{\beta}))} \\ &\quad \times M_i(T_0)^{-1} f(u) du \\ &= -\underline{x}_i \underline{x}_i' M_i(T_0)^{-1} \int_{-\infty}^{\infty} \delta(u - \underline{x}_i'(\underline{b} - \underline{\beta})) \\ &\quad \times e^{\psi_\tau(u - \underline{x}_i'(\underline{b} - \underline{\beta}))} T_0' \underline{x}_i f(u) du \\ &= -\underline{x}_i \underline{x}_i' M_i(T_0)^{-1} e^{\psi_\tau(0)} T_0' \underline{x}_i f(\underline{x}_i'(\underline{b} - \underline{\beta})). \end{aligned} \quad (25)$$

The last step follows from the well-known property of the Dirac function that

$$\int_{-\infty}^{\infty} \delta(x - x_0) h(x) dx = h(x_0)$$

(see e.g. Wilmott et al., 1995, p. 65). In (25) $\psi_\tau(0)$ is shorthand for

$$\psi_\tau(0) = \lim_{u \rightarrow 0} \rho_\tau'(u).$$

Now, approximating ρ'_τ by the function

$$d_\varepsilon(u) = \begin{cases} \tau - 1 & \text{for } u < -\varepsilon \\ \tau & \text{for } u > +\varepsilon \\ \frac{u}{2\varepsilon} + \tau - \frac{1}{2} & \text{for } -\varepsilon \leq u \leq \varepsilon, \end{cases}$$

for small ε , it follows that an appropriate choice of $\psi_\tau(0)$ is $\tau - \frac{1}{2}$. Using this in (25) we obtain

$$E_g \cdot \left[\frac{\partial \psi_i}{\partial \underline{b}} \right] = -\underline{x}_i \underline{x}'_i M_i(\underline{T}_0)^{-1} e^{(\tau - \frac{1}{2})(\underline{T}_0)' \underline{x}_i} f(\underline{x}'_i(\underline{b} - \underline{\beta})). \quad (26)$$

Putting these results together, we obtain our final approximation for the density of $\hat{\beta}_{RQ}$ as

$$\begin{aligned} f_\tau^{(SP)}(\underline{b}) &= (2\pi)^{-p/2} \exp[K(\underline{T}_0)] |K''(\underline{T}_0)|^{-\frac{1}{2}} \\ &\times \left| \sum_{i=1}^n \underline{x}_i \underline{x}'_i M_i(\underline{T}_0)^{-1} f(\underline{x}'_i(\underline{b} - \underline{\beta})) e^{(\tau - \frac{1}{2}) \underline{x}'_i \underline{T}_0} \right|, \quad (27) \end{aligned}$$

with $|\cdot|$ denoting the absolute value of the determinant of the enclosed matrix in the last factor.

Remarks

1. In applying (27) we multiply by a normalizing constant so that $f_\tau^{(SP)}$ integrates to one.
2. Our (27) corresponds closely to result (8) in Strawderman et al. (1996) and to Ronchetti and Welsh (1994), §4. In particular, the matrix in (26) is the same as the matrix $A(v)$ in Ronchetti and Welsh (1994) and the matrix A in Strawderman et al. (1996), p. 645. We found Spady's result slightly more convenient to work with.
3. In the next section we investigate the accuracy of the approximations in (27) and (8), compared to the exact distribution of $\hat{\beta}_{RQ}(\tau)$, for different choices of τ .

4. Simulation Study

In this section we discuss the design of a numerical study that was carried out in order to assess:

- How well the asymptotic normal distribution of the RQs approximates its finite sample distribution;
- The improvement in this approximation one can expect by using the saddlepoint method discussed in the previous section.

For this, three densities were calculated:

- The “true” density, obtained by simulating a large number of RQs, at the different levels of the design variables;
- The appropriate asymptotic normal density, given by (8), and
- The saddlepoint density, given by (27) and normalized by numerical integration.

In order to assess the approximations, two measures were used:

- As a graphical measure, we display the three densities on the same graph for given F, τ, n . In order to highlight the differences, especially in the tails, we also show the logarithms of the densities. To save space, we mostly give the latter.
- As a numerical measure, we used the Hellinger distance between the simulated density and the corresponding approximate density. For two densities f_1 and f_2 , their Hellinger distance is defined as

$$H(f_1, f_2) = \left[\int_{R^P} \left(f_1(\underline{x})^{\frac{1}{2}} - f_2(\underline{x})^{\frac{1}{2}} \right)^2 d\underline{x} \right]^{\frac{1}{2}} \quad (28)$$

(see e.g. Pollard, 2002). Note that this distance can be used for one-dimensional densities as well as for higher dimensional ones. We give these distances for the joint as well as the marginal densities.

The simulation study was designed as follows:

The study was confined to $p = 2$. Without loss of generality we took $\beta_1 = 0, \beta_2 = 1$.

Since $p = 2$, there was only one design variable, say x . Values for x were chosen in such a way that the design matrix X has $X'X = \text{diag}(n, n)$. This is easily obtained by taking the x_i as the standardized values of the numbers $1, 2, \dots, n$.

Three sample sizes were chosen, viz. 15, 20 and 50.

Three choices were made for the underlying distribution:

- A normal distribution with mean zero and variance 1.25. This distribution will be denoted by N .
- A contaminated normal distribution with mean zero, variance 1.25 and coefficient of kurtosis 15.8. This was chosen in order to investigate the effect of heavy tails on the results. The cumulative distribution function is

$$(0.1) \Phi\left(\frac{x}{3}\right) + (0.9) \Phi\left(\frac{x}{0.6236}\right)$$

and will be denoted by CN .

- An equiprobable mixture of two normals with means 1 and -1 and common variance 0.25. This distribution has mean zero and variance 1.25 and is bimodal with a deep valley at zero. For this distribution one might expect the usual asymptotic normality for the RQs to be misleading at small sample sizes. The cumulative distribution function is

$$0.5 \left[\Phi\left(\frac{x-1}{\sqrt{0.25}}\right) + \Phi\left(\frac{x+1}{\sqrt{0.25}}\right) \right]$$

and will be denoted by MN . See Spady (1991) where this distribution was also used.

The (simulated) empirical densities were estimated by using non-parametric kernel density estimation techniques (see e.g. Wand and Jones, 1995). We used the biweight kernel since this has finite support, making numerical evaluation faster.

- The results of the simulation study are given in the next section.

5. Results

In this section we report some of the results obtained in the simulation study. The study was carried out for a range of τ values in the interval $[0.5; 0.9+]$. For τ larger than 0.9, the results were numerically unstable, so we only report $\tau \leq 0.9$. Furthermore, in order to save space, we only give results for $\tau = 0.5, 0.7, 0.9$. These represent central, intermediately large and large (“extreme”) τ values.

5.1 Graphical Results

In order to graphically indicate where deviations from the true distribution occurred, we obtained densities of the normal and saddlepoint approximations. For each factor combination, the three densities are given on the same graph. The asymptotic normal approximation and the saddlepoint approximation can thus be compared to the true (simulated) density. In order to highlight differences, we usually graph the log- densities. The *MN* distribution is the exception since deviations of the asymptotic normal approximation from the true are already very clear by graphing the densities themselves. A selection of the graphs is given as an appendix and a discussion of the results is in section 6. Since our interest is primarily in small samples, we only give the results for $n = 15$. A version of the paper containing a full set of graphs is available as a technical report at www.sun.ac.za/statistics/cvtertius.

5.2 Numerical Results

As an overall quantitative measure of deviation between the approximate densities and the corresponding true ones, we use the Hellinger distance as defined in (28). This can be calculated for the joint as well as the marginal distributions. Since a comparison between the asymptotic and saddlepoint approximations is important, Table 1 below gives the ratios of

Hellinger distance for the asymptotic approximation to that for the saddlepoint approximation. These values should therefore be compared to one: A value larger than one implies that an improvement is obtained by using the saddlepoint method rather than asymptotic normality. In the table, b_1 and b_2 denote respectively the Hellinger ratios for the marginals for intercept and slope and (b_1, b_2) the Hellinger ratios of the joint distributions. These results are also discussed in section 6.

6. Discussion of Results

- 6.1 For both N and CN , the asymptotic normal approximation does (surprisingly) well, even at the small sample size of 15. It does drift away a bit in the tails, especially for larger τ .
- 6.2 The saddlepoint approximation on the other hand, at times, tends to be higher in the centre for N and CN .
- 6.3 Overall, the saddlepoint approximation gives a very good improvement over the asymptotic approximation. The best gain is obtained in estimating the intercept parameter rather than the slope parameter.
- 6.4 For the MN distribution, the asymptotic approximation is poor, while the saddlepoint approximation is still good. Especially at $\tau = 0.5$ the former breaks down. Its relative performance does improve for larger τ . Since MN is a bimodal distribution, one would not expect the asymptotic approximation to be good. For larger τ the effect of the bimodality decreases, as one would expect. However, the effect does not decrease with sample size.
- 6.5 Note from the Hellinger ratios that overall one gains relatively more by using the saddlepoint for N and MN at more central τ values, while for CN this higher gain is at the more extreme τ values.
- 6.6 Note the "bumpy" nature of the simulated density for MN at

$\tau = 0.5$. This bumpiness essentially disappears for larger τ .

6.7 We also experimented with n smaller than 15 (e.g. $n = 10$), but found too much numerical instability for the results to be reported.

6.8 From the densities one can easily obtain moments and cumulants. We do not report these but rather give the Hellinger ratios as overall measures of relative performance.

7. Conclusions and further work

We found the asymptotic normal approximation to perform surprisingly well, even at small sample sizes such as $n = 15$. An exception is in the case of a bimodal type distribution at central τ values. Overall there is improvement by using the saddlepoint approximation, with this improvement more pronounced in estimating the intercept than the slope. Also, the improvement is better when the conditions for asymptotic normality do not hold for the underlying distribution.

Our numerical study was confined to the case $p = 2$ and a well behaved design. Further work is needed to investigate the behaviour of the approximations at larger values of p and/or for less well behaved designs.

In this paper the saddlepoint approximation was applied using a known underlying distribution and regression parameter values. In practice this is of course not the case and they have to be estimated. Work is in progress on evaluating an empirical saddlepoint approximation and comparing it to, e.g. a bootstrap approximation (see Ronchetti and Welsh, 1994 in this regard). These results will be reported in a subsequent paper.

Table 1. Hellinger ratios for $n = 15, 20, 50$

$n = 15$		(b_1, b_2)	b_1	b_2
	tau			
$F = N$	0.5	3.71	3.90	5.08
	0.7	2.93	2.60	4.21
	0.9	2.41	1.78	2.50
$F = CN$	0.5	1.40	1.12	1.88
	0.7	2.80	2.94	3.51
	0.9	5.06	3.15	4.63
$F = MN$	0.5	19.02	24.81	27.14
	0.7	8.10	9.07	6.72
	0.9	2.58	2.34	3.30
$n = 20$		(b_1, b_2)	b_1	b_2
	tau			
$F = N$	0.5	3.53	8.08	2.96
	0.7	2.92	7.97	2.21
	0.9	2.28	6.38	0.76
$F = CN$	0.5	1.12	1.53	1.01
	0.7	2.36	4.39	1.84
	0.9	5.19	4.59	4.26
$F = MN$	0.5	15.93	21.06	13.39
	0.7	7.83	14.39	5.21
	0.9	2.38	7.19	1.99
$n = 50$		(b_1, b_2)	b_1	b_2
	tau			
$F = N$	0.5	4.21	15.32	6.72
	0.7	4.38	16.08	5.91
	0.9	3.63	11.35	1.97
$F = CN$	0.5	0.92	1.05	0.88
	0.7	1.46	4.55	1.33
	0.9	7.11	11.25	6.86
$F = MN$	0.5	23.36	86.54	26.79
	0.7	8.39	28.56	9.57
	0.9	1.93	7.85	1.62

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APPENDIX: GRAPHICAL RESULTS

Simulated = _____ Saddlepoint = Asymptotic = _ . _ . _ . _

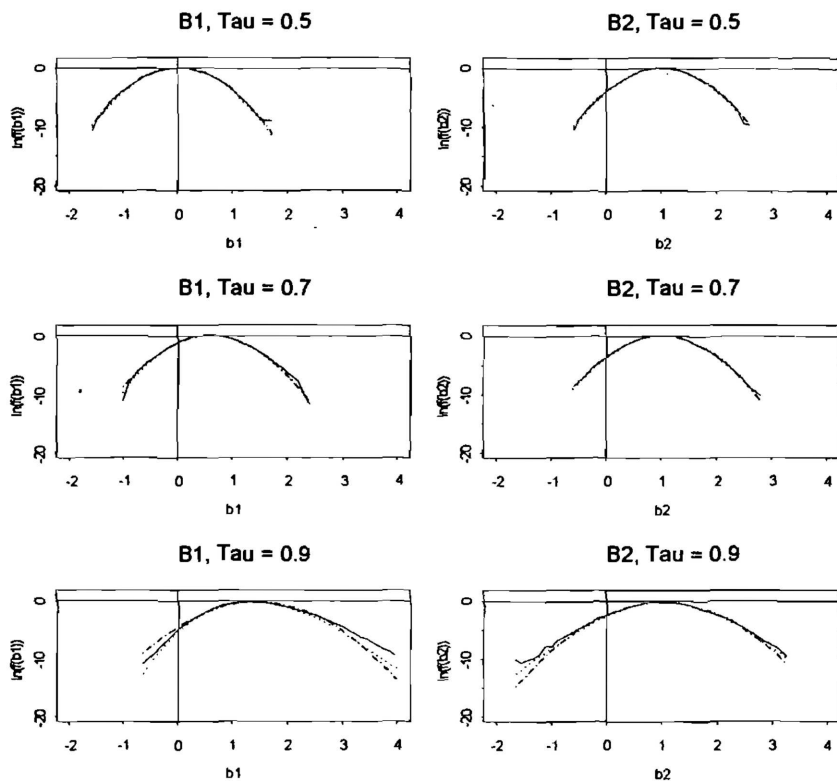


Figure 1. Log densities for normal underlying distribution and sample size 15, $\tau = 0.5, 0.7, 0.9$

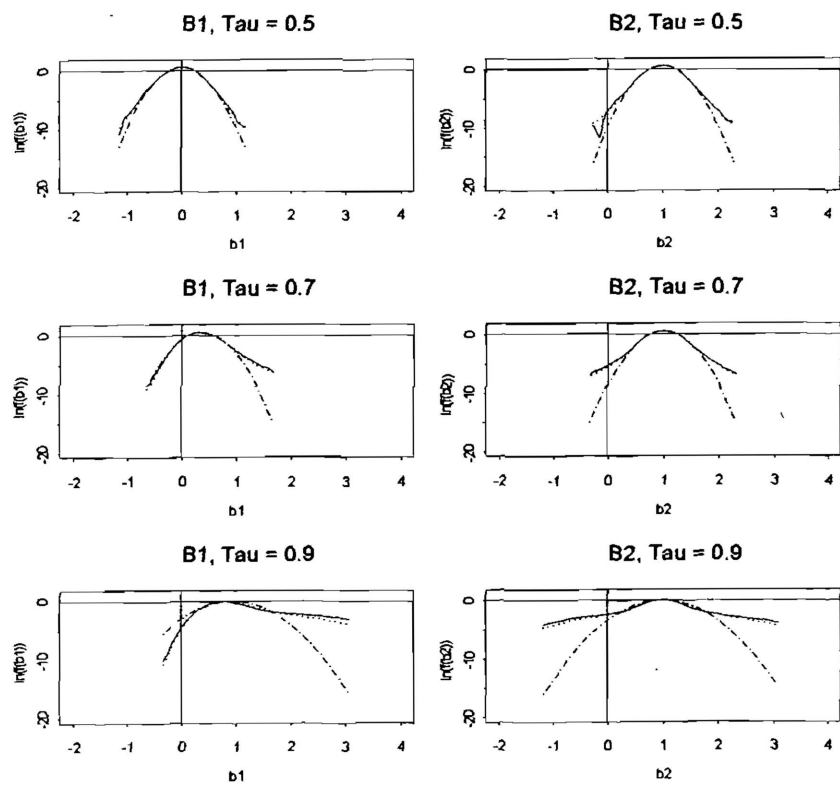


Figure 2. Log densities for contaminated normal underlying distribution and sample size 15, $\tau = 0.5, 0.7, 0.9$

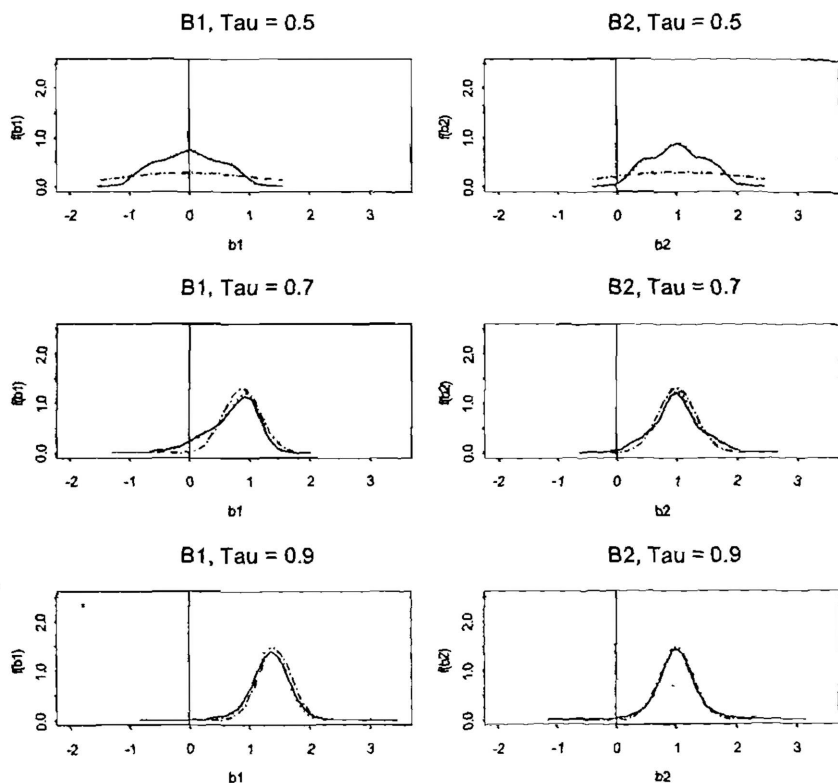


Figure 3. Densities for mixed normal underlying distribution and sample size 15, $\tau = 0.5, 0.7, 0.9$